WSQ COURSE SLIDES

Data Visualisation with Tableau

Course Code: TGS-2020503177

Data Visualization · ICT-DIT-3006-1.1

Trainer: Dr Alfred Ang

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v17.0 · 15 August 2026

Concept visual generated for this course

00

COURSE ADMINISTRATION

Welcome and Course Orientation

Digital Attendance (Mandatory)

1

AM attendance
Complete SSG digital attendance at the start of each day.

2

PM attendance
Complete the second attendance at the instructed time.

3

Assessment attendance
Take the separate digital attendance before the papers.

4

Eligibility
Maintain at least 75% attendance and pass the assessment.

About the Trainer

?

General Trainer

Complete this profile before delivery

NAME

TITLE / DESIGNATION

QUALIFICATIONS

EXPERTISE

EXPERIENCE

CONTACT

NAMED TRAINER

About the Trainer

AA

Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE	Principal Trainer and Chief Instructional Designer
EXPERTISE	Data visualisation, analytics, AI and business reporting
CERTIFICATION	ACLP-certified adult educator
DELIVERS	WSQ and professional technology programmes
EXPERIENCE	Cross-industry analytics and workforce training
CONTACT	enquiry@tertiaryinfotech.com

Let's Know Each Other

1 **Your role**
What decisions do you support?

2 **Your Tableau baseline**
What have you already built?

3 **Your data challenge**
Where does interpretation break down?

4 **Your outcome**
What visual decision product do you want to create?

HOUSEKEEPING

Ground Rules

1 **Be present**
Return on time and complete digital attendance.

2 **Participate**
Ask, test and explain your reasoning.

3 **Respect**
Critique ideas, not people.

4 **Protect data**
Use only the supplied synthetic datasets.

5 **Save versions**
Keep source, work-in-progress and packaged evidence.

6 **Learn by doing**
Attempt before opening the sample workbook.

SCHEDULE

Lesson Plan — Two Days

DAY 1

1. 9:00–10:30 · Tableau foundations
2. 10:40–12:30 · Labs 1–2
3. 1:30–3:15 · Sources and chart choice
4. 3:25–6:00 · Labs 3–5

DAY 2

1. 9:00–10:30 · Dashboards and actions
2. 10:40–12:30 · Labs 6–8
3. 1:30–4:00 · Calculations, analytics and Labs 9–10
4. 4:00–6:00 · WA and PP assessment

WHAT YOU WILL ACHIEVE

Learning Outcomes

1

LO1
Apply basic Tableau features for displaying information.

2

LO2
Synthesise various plots to present data visually.

3

LO3
Transform data to create informative and dynamic data displays.

4

LO4
Create dashboards and scorecards to display internal and external benchmark data.

5

LO5
Apply calculations and parameters to create interactive graphics, visuals and technical features.

6

LO6
Perform analytics and communicate data limitations and interpretations of findings.

Skills Framework Alignment

1 **K1**
Interpretation of data analysis and findings

2 **K2**
Types of information displays

3 **K3**
Suitability of data representations and visual displays for different contexts

4 **K4**
Data visualisation tools and techniques

5 **K5**
Elements of data dashboards

6 **A1–A6**
Practical abilities span technique selection, visual organisation, dynamic displays, dashboards, interactivity and responsible interpretation.

Briefing for Assessment

1. **Individual**
   No discussion or sharing during either paper.

2. **Open book**
   Use slides, Learner Guide and approved materials only.

3. **Devices**
   Place phones and unrelated materials under the table.

4. **Evidence**
   Submit your own Tableau workbook and screenshots.

5. **Integrity**
   No photos, recording or verbatim copying.

6. **Process**
   Re-assessment and appeal arrangements are available.

Assessment and Funding

Written Assessment

- 5 open-ended questions
- 60 minutes · open book

Practical Performance

- 6 connected Tableau tasks
- 60 minutes · open book

Outcome

- Competent or Not Yet Competent
- Attendance and submission requirements apply

Assessment Flow

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then PP
4. Submit on the LMS
5. Sign Assessment Summary Record

Access the Hands-On Labs

Option A · Clone

- Clone the GitHub repository
- Open the numbered lab folder
- Start with the root Excel file

Option B · Download ZIP

- GitHub > Code > Download ZIP
- Extract the full folder tree
- Keep Tableau and Excel together

Lab discipline

- Use the LG for detailed procedure
- Reconcile against Excel controls
- Package evidence as .twbx

https://github.com/tertiarycourses/TGS-2020503177-Data-Visualisation-with-Tableau

Download Course Material

1 · Sign in

- Open the LMS/TMS
- Use your learner account
- Select this course

2 · Download

- Learner slides and guide
- Keep the lab folders intact
- Use the current version

https://lms-tms.tertiaryinfotech.com/

3 · Verify

- Check TGS-2020503177
- Confirm version 17.0
- Ask the trainer if files differ

BEFORE HANDS-ON

Tableau Safety and File Discipline

1

Public platform

Tableau Public is public by design; use only supplied synthetic data.

2

Packaged evidence

Save learner work as .twbx so local files travel with the workbook.

3

Source preservation

Keep the supplied Excel source unchanged and reconcile against its Control sheet.

4

Responsible claims

State filter scope, assumptions, limitations and monitoring trigger.

From Source to Decision Story

LEARNING JOURNEY · 6 TOPICS

6

Connect → Model → Visualise → Interact → Analyse → Communicate

The six linked topics progress from reliable source connection to responsible decision communication; detailed procedures stay in the Learner Guide.

01

TOPIC 01

Basic Tableau Features

Tableau ecosystem · interface · data roles · shelves · hierarchy

ALIGNMENT K1 · A1 · LO1

Key Concepts — Basic Tableau Features

1 **Visual analysis cycle**
Connect, explore, explain and refine around a decision question.

2 **Dimensions and measures**
Dimensions slice the view; measures are aggregated to quantify it.

3 **Discrete and continuous**
Blue headers create categories; green axes create a continuous range.

4 **Workbook structure**
Worksheets build views; dashboards combine them; stories sequence insights.

5 **Fields and metadata**
Names, types, roles, aliases and default properties shape every view.

6 **Hierarchy**
Ordered levels let users drill from overview to operational detail.

What Tableau Adds to Decision Work

Fast exploration

- Drag-and-drop visual analysis
- Immediate feedback on questions

Connected evidence

- Files, databases and governed sources
- Live or extracted data

Communicable output

- Worksheets, dashboards and stories
- Publish with context and controls

The Visual Analysis Cycle

1. Frame the decision question
2. Connect and profile data
3. Build and compare views
4. Interpret patterns and limits
5. Share and monitor action

Tableau Products and Roles

CONCEPT

1. Desktop / Public author views and workbooks.
2. Cloud / Server govern sharing and collaboration.
3. Prep supports repeatable cleaning and reshaping.
4. Reader opens packaged workbooks offline.

DECISION LENS

1. Choose by data sensitivity and audience.
2. Tableau Public is public by design.
3. Publishing permissions are part of analysis quality.
4. The course uses Desktop or Public for authoring.

Workspace: One Canvas, Several Control Surfaces

1 **Data pane**
Fields, folders, hierarchies and metadata.

2 **Shelves**
Rows, Columns, Pages and Filters shape structure.

3 **Marks card**
Colour, size, label, detail, tooltip and shape.

4 **View**
Marks, axes, headers, legends and annotations.

Dimensions and Measures

DIMENSIONS

1. Dimensions describe who, what, where and when.
2. They create headers and define the level of detail.
3. Examples: Region, Category, Customer and Order Date.

MEASURES

1. Measures quantify how much or how many.
2. They are usually aggregated in the view.
3. Examples: Sales, Profit, Quantity and Resolution Hours.

Discrete and Continuous

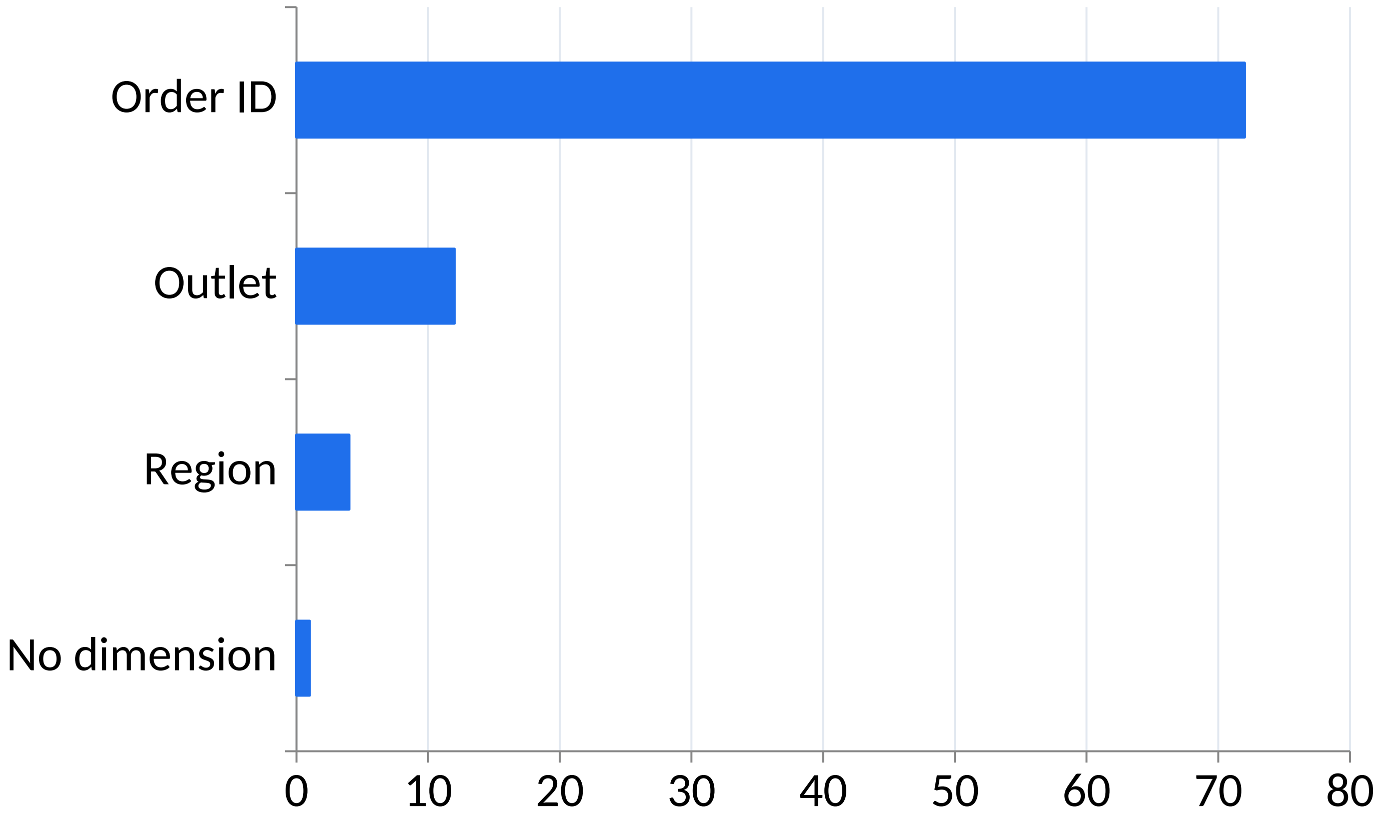
1. **Discrete field**
   Creates separate headers; shown as a blue pill.

2. **Continuous field**
   Creates an axis across a range; shown as a green pill.

3. **Date choice**
   Year may be discrete while exact date can be continuous.

4. **Design effect**
   The same source field can create a different visual structure.

Measures Aggregate at the View Level



READ THE EVIDENCE

Adding dimensions increases the number of marks and changes the level at which a measure is aggregated.

Editable chart: change the embedded data, labels, axes and colours directly in PowerPoint.

Rows, Columns and the Level of Detail

1 **Rows**
Positions headers or axes vertically.

2 **Columns**
Positions headers or axes horizontally.

3 **Detail**
Adds marks without necessarily adding a visible encoding.

4 **Granularity**
Every added dimension can change the number of marks.

Marks Encode Meaning

Position and length

- Strong for accurate comparison
- Preferred for core quantitative questions

Colour and size

- Useful for emphasis and additional variables
- Need accessible legends and restraint

Label and tooltip

- Add exact values and context
- Should not compensate for a weak view

Workbook Structure

1. Data source
2. Worksheet
3. Dashboard
4. Story
5. Published workbook

File Types and Portability

1 **.twb**
XML workbook definition; external data remains separate.

2 **.twbx**
Packaged workbook with local data and assets.

3 **.hyper**
Tableau extract optimised for analytical queries.

4 **.tds / .tdsx**
Reusable data-source metadata, optionally packaged.

Folders and Hierarchies

FOLDERS

1. Folders organise the authoring experience.
2. They do not change the source data or aggregation.
3. Use names that reflect business domains.

HIERARCHIES

1. Hierarchies define an ordered drill path.
2. Typical paths: Region > District > Outlet.
3. Every level should answer a meaningful next question.

Connect and Profile Service Operations Data

LAB 1

A customer-experience manager needs a first reliable view of service tickets across channels and regions.

EXPECTED OUTPUT

A connected Tableau workbook with validated field roles and a service-volume view by region and channel.

TOOLS Tableau Desktop/Public · Data Source page · Dimensions/Measures · Show Me

Lab 1 Acceptance Evidence

ACCEPTANCE TEST

The workbook connects without broken fields; dates and numbers have correct roles; the view reconciles to the Excel record count.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Build Hierarchies and Drill Paths

LAB 2

Retail leaders need to move from national performance to region, outlet and department detail without losing context.

EXPECTED OUTPUT

A geographic and product hierarchy with a drillable sales-and-margin view.

TOOLS Folders · Hierarchies · Drill up/down · aliases · default properties

Lab 2 Acceptance Evidence

ACCEPTANCE TEST

Both hierarchies drill in the correct order and totals remain stable at every displayed level.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

TOPIC RECAP

Recap — Basic Tableau Features

1 **Knowledge**
Visual analysis cycle and related principles.

2 **Ability**
K1 · A1 · LO1

3 **Evidence**
A packaged Tableau workbook plus reconciliation and interpretation.

4 **Next**
Connect this layer to the next decision requirement.

02

TOPIC 02

Data Sources and Visualisation

Connections · extracts · relationships · joins · blending · chart choice

ALIGNMENT K3 · A2 · LO2

Key Concepts — Data Sources and Visualisation

1 **Connection strategy**
Choose live or extract by freshness, performance and governance needs.

2 **Relationships**
Preserve each table's grain and combine data at analysis time.

3 **Joins**
Physically combine rows; cardinality and unmatched keys determine the result.

4 **Visual encoding**
Position and length usually support more accurate comparisons than area or colour.

5 **Chart purpose**
Use bars for comparison, lines for time, scatterplots for relationship and treemaps for hierarchy.

6 **Aggregation**
SUM, AVG, COUNT and distinct count answer different business questions.

Live Connection or Extract?

LIVE

1. Queries the source at interaction time.
2. Best when current values and central controls matter.
3. Performance depends on the source and network.

EXTRACT

1. Stores a local analytical snapshot.
2. Supports fast interaction and offline work.
3. Requires a refresh policy and timestamp.

Connection Readiness Gate

1. Confirm business grain
2. Profile keys and types
3. Choose live or extract
4. Model tables safely
5. Reconcile totals

Three Ways to Combine Data

Relationships

- Logical layer
- Preserve table grain

Joins

- Physical row combination
- Cardinality changes rows

Blending

- Separate data sources
- Aggregate secondary results

Join Types and Their Consequences

1 **Inner**
Keeps only matched rows.

2 **Left**
Keeps every left row plus matching right data.

3 **Right**
Keeps every right row plus matching left data.

4 **Full outer**
Keeps matched and unmatched rows from both sides.

Relationships versus Joins

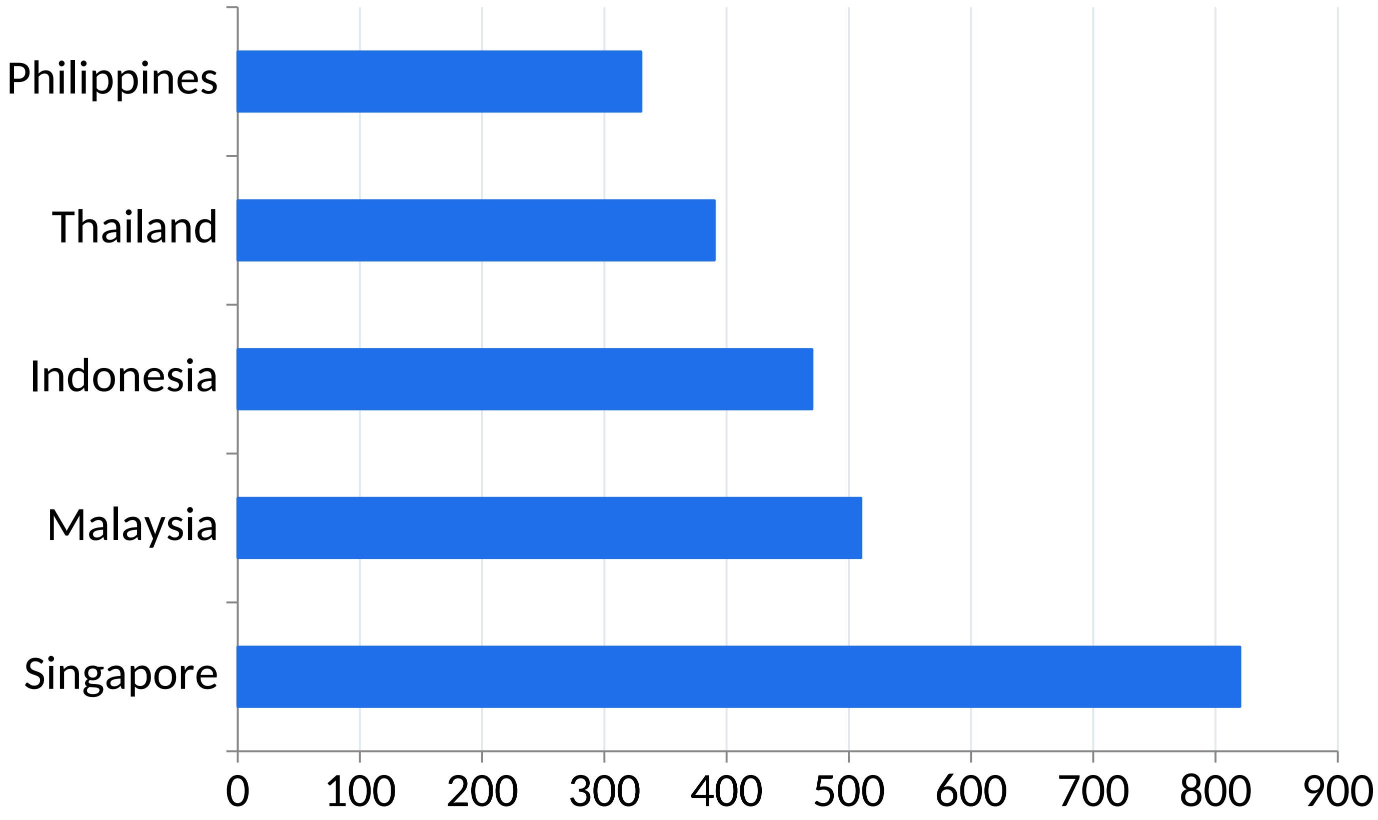
LOGICAL MODEL

1. Relationships are resolved in the context of the view.
2. They reduce accidental duplication across different grains.
3. Use defined cardinality and referential integrity when known.

PHYSICAL MODEL

1. Joins create one physical row set before analysis.
2. One-to-many matches can duplicate measures.
3. Use joins when row-level combination is truly required.

Ranked Comparison: Sales by Market

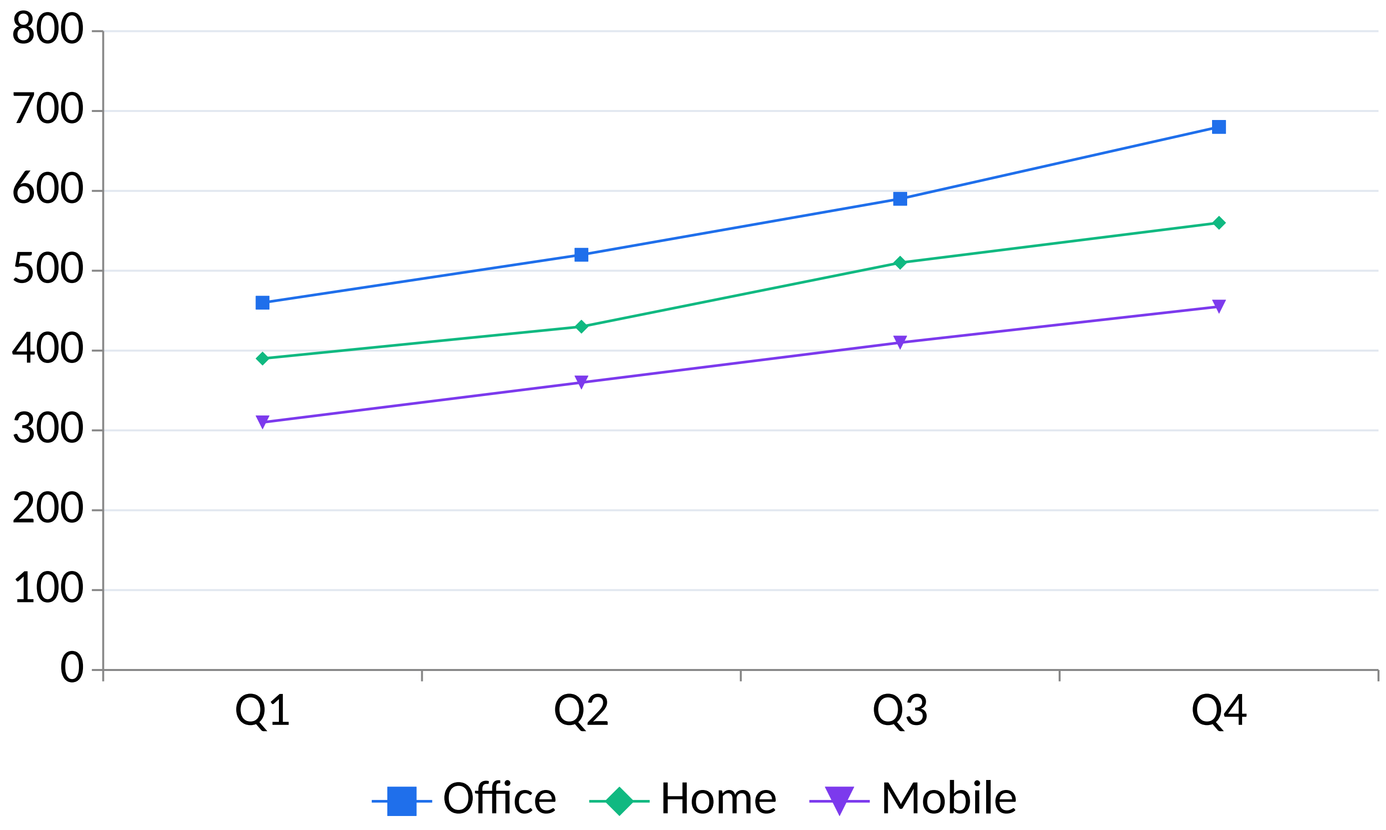


READ THE EVIDENCE

Singapore leads the sample, but regional budget allocation should also consider margin and growth.

Editable chart: change the embedded data, labels, axes and colours directly in PowerPoint.

Trend: Quarterly Sales by Category



READ THE EVIDENCE

All categories rise in Q4; Office contributes the largest absolute increase.

Editable chart: change the embedded data, labels, axes and colours directly in PowerPoint.

Relationship: Sales and Profit

Pattern

- Marks generally rise from lower-left to upper-right
- Higher sales are associated with higher profit in the sample

Exception

- One high-sales outlet has weaker profit
- Use Outlet on Detail to identify it

Caution

- Association does not establish causality
- Inspect discount, mix, cost and data quality

Choose a Chart from the Question

1 **Comparison**
Bar or dot plot.

2 **Time**
Line or area chart with an ordered axis.

3 **Relationship**
Scatterplot with a meaningful level of detail.

4 **Distribution**
Histogram or box plot.

5 **Composition**
Stacked bar or treemap when part-to-whole matters.

6 **Location**
Map only when geography changes the decision.

Portfolio Views

Treemap

- Hierarchical part-to-whole
- Area is less precise than length

Heatmap

- Pattern across two categorical axes
- Colour scale needs clear meaning

Box plot

- Median, spread and unusual values
- Good for group distributions

Aggregation Changes the Question

CALCULATION

1. SUM answers total contribution.
2. AVG answers the typical row at the current grain.
3. COUNT counts records; COUNTD counts unique entities.
4. ATTR signals whether a single value is present.

CONTROL

1. Confirm the denominator behind every average.
2. Do not average percentages when a weighted ratio is needed.
3. Display the level of detail in titles or captions.
4. Reconcile to a known control total.

TOPIC 02 · K3 · A2 · LO2

A view is only as trustworthy as its grain.

Before formatting a chart, prove what one row means and whether combining tables changes the population.

LAB 3 · LO2 · K3 · A2

Relate Orders, Products and Returns

LAB 3

Meridian Retail SG stores orders, product master data, outlets and returns at different grains.

EXPECTED OUTPUT

A governed multi-table data model plus an extract and reconciliation sheet.

TOOLS Relationships · joins · cardinality · referential integrity · extract

Lab 3 Acceptance Evidence

ACCEPTANCE TEST

Order totals are not duplicated, unmatched keys are visible, and the extract refreshes from the supplied workbook.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Choose and Build Decision Charts

LAB 4

A commercial director needs comparison, trend, relationship and portfolio views for one monthly review.

EXPECTED OUTPUT

Four fit-for-purpose worksheets: ranked bar, quarterly line, profit-sales scatterplot and category treemap.

TOOLS Bar · line · scatterplot · treemap · labels · tooltips · sort

Lab 4 Acceptance Evidence

ACCEPTANCE TEST

Every chart answers its stated question, uses truthful axes, and includes enough context to interpret the marks.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

TOPIC RECAP

Recap — Data Sources and Visualisation

1 **Knowledge**
Connection strategy and related principles.

2 **Ability**
K3 · A2 · LO2

3 **Evidence**
A packaged Tableau workbook plus reconciliation and interpretation.

4 **Next**
Connect this layer to the next decision requirement.

03

TOPIC 03

Data Transformation

Data Interpreter · split · merge · pivot · filters · groups · sets

ALIGNMENT K4 · A3 · LO3

Key Concepts — Data Transformation

1. **Analysis-ready structure**
   One row is one observation; one column is one variable; one table is one grain.

2. **Data Interpreter**
   Removes presentation clutter from spreadsheets while preserving the underlying evidence.

3. **Split and pivot**
   Separate compound fields and reshape wide columns into analysis-ready rows.

4. **Filter order**
   Extract, source, context, dimension and measure filters do not behave identically.

5. **Groups**
   Combine members into a stable business category.

6. **Sets**
   Define a dynamic in/out population that can drive comparisons and actions.

From Presentation Sheet to Analytical Table

1. Detect the real header
2. Remove presentation clutter
3. Separate compound fields
4. Pivot repeated columns
5. Validate the long table

What Data Interpreter Can and Cannot Do

AUTOMATION

1. Detects headers, notes, blank rows and repeated formatting.
2. Helps expose a clean table from a human-formatted sheet.
3. Shows what was removed for review.

JUDGEMENT

1. It cannot decide the business grain.
2. It does not fix ambiguous definitions.
3. A clean-looking result still needs reconciliation.

Split, Merge and Aliases

1 **Split**
Turn Region|District into separate fields.

2 **Custom split**
Choose delimiter and occurrence deliberately.

3 **Merge**
Combine fields only when the new meaning is clear.

4 **Aliases**
Change display labels without changing source values.

Wide versus Long Data

WIDE

1. One row may contain Jan, Feb, Mar columns.
2. Easy for manual reading; awkward for time analysis.
3. New months require new fields.

LONG

1. Month becomes a field and value becomes a measure.
2. Works naturally with time axes and filters.
3. New months append new rows.

Filter Order of Operations

1. Extract filter
2. Data source filter
3. Context filter
4. Dimension filter
5. Measure / table-calc filter

Dimension Filters

1 **General**
Select explicit members.

2 **Wildcard**
Match text patterns.

3 **Condition**
Retain members that meet a rule.

4 **Top**
Keep leading members by an aggregated measure.

Measure and Date Filters

1. **Range of values**
   Filter aggregated numeric results.

2. **At least / at most**
   Apply a clear materiality threshold.

3. **Relative date**
   Rolling periods that update with time.

4. **Range of dates**
   Fixed window for controlled comparisons.

Context Filters

PURPOSE

1. Create a temporary population for dependent filters.
2. Useful when Top N should apply within a selected region.
3. Can improve or reduce performance depending on design.

RISK

1. Do not use context as a visual-formatting trick.
2. Explain the population behind the result.
3. Test whether changing context changes set membership.

Groups, Sets and Bins

Group

- Stable business consolidation
- Manual or data-driven membership

Set

- Dynamic IN / OUT population
- Supports comparison and actions

Bin

- Intervals for a continuous measure
- Sensitive to bin width

Set Design Patterns

1 **Top N**
Leading customers or products within context.

2 **Condition**
Members above a measure threshold.

3 **Combined set**
Intersection or union of two populations.

4 **Set action**
Let dashboard interaction change membership.

Transformation Controls

RECONCILE

1. Record count before and after.
2. Key completeness and uniqueness.
3. Total spend, sales or units.
4. Explicit list of excluded records.

GOVERN

1. Retain the raw source.
2. Document every transformation.
3. Separate correction from assumption.
4. Make limitations visible to the audience.

TOPIC 03 · K4 · A3 · LO3

Cleaning is a chain of evidence, not a cosmetic exercise.

Every transformation should preserve meaning, expose exceptions and be reproducible on refresh.

Clean, Split and Pivot Campaign Data

LAB 5

A campaign export contains banner rows, compound location codes and monthly measures spread across columns.

EXPECTED OUTPUT

An analysis-ready long table with clean location fields and documented transformation checks.

TOOLS Data Interpreter · custom split · merge · pivot · aliases

Lab 5 Acceptance Evidence

ACCEPTANCE TEST

The transformed data has one row per campaign-month, clean location fields, no blank campaign key and reconciled spend totals.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

LAB 6 · LO3 · K4 · A3

Filter, Group and Set Customer Segments

LAB 6

The loyalty team wants a stable business grouping plus a dynamic high-value customer cohort.

EXPECTED OUTPUT

A segment view using context filters, grouped categories and a top-customer set.

TOOLS Dimension/measure/date filters · context filter · groups · sets

Lab 6 Acceptance Evidence

ACCEPTANCE TEST

The high-value set responds to the selected period and the business grouping is explainable from the source fields.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

TOPIC RECAP

Recap — Data Transformation

1 **Knowledge**
Analysis-ready structure and related principles.

2 **Ability**
K4 · A3 · LO3

3 **Evidence**
A packaged Tableau workbook plus reconciliation and interpretation.

4 **Next**
Connect this layer to the next decision requirement.

Lunch Break

45 minutes

04

TOPIC 04

Dashboards and Stories

KPI scorecards · layout · actions · device design · story sequence

Key Concepts — Dashboards and Stories

1 **Four Ws**
Who is the audience, what decision, when is it used and where is action taken?

2 **Dashboard hierarchy**
Lead with decision KPIs, then trends, drivers and diagnostic detail.

3 **Layout containers**
Tiled containers create predictable alignment; floating objects need deliberate control.

4 **Actions**
Filter, highlight, URL, navigation, parameter and set actions create purposeful interaction.

5 **Benchmark context**
Targets, prior periods and peer ranges turn values into performance signals.

6 **Stories**
A sequence of story points communicates change, evidence and decision.

TOPIC 04 · K5 · A4 · LO4

The Four Ws of a Good Dashboard

1 **Who**
Audience, authority and data literacy.

2 **What**
Decision, question and required action.

3 **When**
Refresh cadence and time pressure.

4 **Where**
Presentation, desktop or mobile context.

Dashboard Information Hierarchy

1. Decision KPIs ›
2. Trend and benchmark ›
3. Drivers and segments ›
4. Diagnostic detail ›
5. Action and ownership

KPI Value versus KPI Context

VALUE

1. Value alone states only the current magnitude.
2. Large type improves scanning.
3. Consistent units and rounding reduce cognitive load.

CONTEXT

1. Target, prior period and peer benchmark create meaning.
2. Direction should be labelled, not only coloured.
3. Variance needs an owner and threshold.

Layout Choices

Tiled

- Predictable alignment
- Responsive container structure

Floating

- Precise placement
- Higher overlap and maintenance risk

Device layout

- Adapt hierarchy to the screen
- Do not merely shrink desktop

Dashboard Objects with a Job

1. **Horizontal / vertical container**
   Structure and alignment.
2. **Text**
   Context, definitions and action.
3. **Image**
   Brand or orientation, used sparingly.
4. **Blank**
   Intentional spacing, not decoration.
5. **Web page**
   Live reference with security and availability risks.
6. **Navigation**
   Move between summary and detail.

Action Design

1. User intent
2. Source sheet
3. Trigger event
4. Target sheet / URL
5. Reset and feedback

Core Dashboard Actions

Filter

- Narrow target views
- Show selected scope

Highlight

- Preserve context
- Emphasise related marks

URL / navigation

- Open supporting detail
- Validate destination and privacy

Advanced Interaction

1. **Parameter action**
   Write a selected value into a parameter.

2. **Set action**
   Change set membership from marks.

3. **Change parameter**
   Switch metric, threshold or view logic.

4. **Go to sheet**
   Move from overview to diagnostic detail.

Dashboard Performance

DESIGN

1. Limit marks and expensive calculations.
2. Use extracts and filters where appropriate.
3. Reduce redundant worksheets and high-cardinality quick filters.

TEST

1. Measure load and interaction time.
2. Performance is part of usability.
3. Optimise only after validating the business question.

Story Point Sequence

1. Situation
2. Signal
3. Exception
4. Driver
5. Decision and next check

Dashboard QA

1. **Accuracy**
   Values reconcile to source controls.

2. **Clarity**
   Hierarchy is readable at intended size.

3. **Interaction**
   Actions work and reset predictably.

4. **Context**
   Date, unit, filters and definitions are visible.

5. **Accessibility**
   Colour is not the only cue.

6. **Actionability**
   Owner, threshold and next step are explicit.

TOPIC 04 · K5 · A4 · LO4

Interactivity should reduce uncertainty, not create a puzzle.

Every control needs a clear purpose, visible state and predictable reset path.

LAB 7 · LO4 · K5 · A4

Build an Executive Performance Dashboard

LAB 7

Executives need one screen to monitor sales, margin, returns and target attainment across outlets.

EXPECTED OUTPUT

A 16:9 scorecard dashboard with KPI tiles, trend, ranked outlets, benchmark context and visible filter state.

TOOLS Dashboards · layout containers · KPI sheets · device preview · captions

Lab 7 Acceptance Evidence

ACCEPTANCE TEST

The dashboard is readable at presentation size, totals reconcile, active filters are visible and every element supports a named decision.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

Add Actions and Tell a Performance Story

LAB 8

Regional managers need to explore a dashboard and then present a concise evidence sequence to leadership.

EXPECTED OUTPUT

A dashboard with filter/highlight/navigation actions and a five-point performance story.

TOOLS Filter action · highlight action · navigation · URL action · story points

Lab 8 Acceptance Evidence

ACCEPTANCE TEST

Actions work from the intended source sheets, preserve context and the story ends with an evidence-backed action.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

TOPIC RECAP

Recap — Dashboards and Stories

1 **Knowledge**
Four Ws and related principles.

2 **Ability**
K5 · A4 · LO4

3 **Evidence**
A packaged Tableau workbook plus reconciliation and interpretation.

4 **Next**
Connect this layer to the next decision requirement.

05

TOPIC 05

Calculations and Parameters

Calculated fields · table calculations · logic · parameters · dynamic views

ALIGNMENT K2 · A5 · LO5

Key Concepts — Calculations and Parameters

1. **Row-level calculation**
   Evaluated for each record before aggregation.

2. **Aggregate calculation**
   Combines measures at the view's level of detail.

3. **Table calculation**
   Computed over the marks already returned to the visualisation.

4. **Logical design**
   IF, IIF and CASE translate business rules into transparent classifications.

5. **Parameters**
   Single values controlled by the user; pair them with calculations or actions to change behaviour.

6. **Dynamic view**
   A parameter can switch metrics, thresholds, reference values or dimensions.

Three Calculation Levels

Row level

- Evaluated for each source record
- Examples: Sales - Cost

Aggregate

- Evaluated after grouping
- Examples: SUM(Profit)/SUM(Sales)

Table calculation

- Evaluated over marks in the view
- Examples: percent of total, running sum

Aggregate or Non-Aggregate?

MECHANICS

1. A row-level field can be aggregated later.
2. Mixing aggregate and non-aggregate expressions causes an error.
3. The view grain determines the final aggregation.

DECISION

1. Decide whether the business rule applies per record or per displayed group.
2. Use consistent aggregation inside the expression.
3. Validate with a known small example.

Calculation Function Families

1 **Number**
ROUND, ABS, CEILING and mathematical operations.

2 **String**
LEFT, SPLIT, CONTAINS and case functions.

3 **Date**
DATETRUNC, DATEPART, DATEDIFF and TODAY.

4 **Logical**
IF, IIF, CASE, AND and OR.

5 **Aggregate**
SUM, AVG, COUNTD, MIN and MAX.

6 **Table calculation**
LOOKUP, WINDOW_SUM, RANK and running functions.

IF, IIF and CASE

SYNTAX

1. IF supports multiple conditions and branches.
2. IIF is concise for one true/false decision.
3. CASE compares one expression to listed values.

CONTROL

1. Keep thresholds in parameters or governed fields.
2. Always include an explicit else path.
3. Test blanks, zero and boundary cases.

Table Calculation Addressing

1. Partition the population
2. Choose addressing direction
3. Compute over the marks
4. Check sort order
5. Test filter interaction

Parameter Anatomy

1. **Name**
   Business-facing description.

2. **Data type**
   Integer, real, string, Boolean, date or date-time.

3. **Allowable values**
   All, list or range.

4. **Current value**
   Single state used by calculations or filters.

5. **Control**
   Visible interface for user input.

6. **Action**
   Dashboard marks can update it.

Threshold Parameter Pattern

PATTERN

1. User chooses a materiality threshold.
2. Calculation compares aggregated performance to the parameter.
3. Filter retains the TRUE result.

GOVERNANCE

1. Show the current threshold in title or caption.
2. Choose a defensible default and range.
3. Avoid implying the threshold is data-driven when it is policy.

Dynamic Metric Pattern

Parameter

- Sales, Profit, Margin or Orders
- One selected value

CASE calculation

- Returns the matching measure
- Uses consistent aggregation

Dynamic labels

- Title and tooltip state the metric
- Formatting matches unit

Useful Table Calculations

1 **Percent of total**
Contribution within the partition.

2 **Running total**
Cumulative performance over an ordered dimension.

3 **Difference from**
Change against prior mark or benchmark.

4 **Rank**
Relative position under the current filter state.

Reference Lines and Parameters

CONTEXT

1. Reference lines provide average, median, constant or computed context.
2. Parameters can supply a user-controlled benchmark.
3. Bands express an acceptable range.

CAUTION

1. Label what the reference means.
2. Avoid treating an average as a target.
3. Keep the scale and filter state visible.

Calculation Validation

1. **Small sample**
   Manually verify several records or marks.

2. **Control total**
   Reconcile numerator, denominator and aggregate.

3. **Edge cases**
   Test zero, null, negative and threshold boundaries.

4. **Filter state**
   Check whether calculation order changes the result.

5. **Formatting**
   Match percentages, currency and units.

6. **Explanation**
   A reviewer can restate the rule in plain language.

TOPIC 05 · K2 · A5 · LO5

A dynamic view still needs a stable definition.

Parameters make analysis interactive; they do not replace governed metric logic and visible assumptions.

Create Dynamic Profitability Analysis

LAB 9

Finance wants to switch between sales, profit and margin, and test which products exceed a user-defined threshold.

EXPECTED OUTPUT

A dynamic metric view with calculated fields, parameter controls, table calculations and a threshold filter.

TOOLS Calculated fields · IF/CASE · table calculations · parameters · dynamic titles

Lab 9 Acceptance Evidence

ACCEPTANCE TEST

Metric and threshold controls change the view correctly, formulas reconcile to Excel controls and invalid denominator cases are handled.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

TOPIC RECAP

Recap — Calculations and Parameters

1 **Knowledge**
Row-level calculation and related principles.

2 **Ability**
K2 · A5 · LO5

3 **Evidence**
A packaged Tableau workbook plus reconciliation and interpretation.

4 **Next**
Connect this layer to the next decision requirement.

06

TOPIC 06

Analytics and Communication

Reference lines · trends · forecasts · clusters · limitations · data stories

ALIGNMENT K1 · A6 · LO6

Key Concepts — Analytics and Communication

1. **Reference context**
   Average, median, target and distribution bands help the audience judge performance.

2. **Trend models**
   A fitted line summarises association; it does not establish causality.

3. **Forecast**
   Tableau extends a time series under model assumptions and expresses uncertainty.

4. **Clustering**
   Groups similar marks from selected features; business labels require interpretation.

5. **Limitations**
   Data coverage, quality, granularity, bias and model assumptions bound the conclusion.

6. **Responsible narrative**
   Separate observation, interpretation, recommendation and uncertainty.

Analytics Pane: Context and Models

1 **Reference line**
Constant, average, median or computed benchmark.

2 **Reference band**
Range between values or distribution limits.

3 **Trend line**
Fitted relationship across marks.

4 **Forecast**
Extension of a time series with uncertainty.

5 **Cluster**
Algorithmic grouping from selected measures.

6 **Box plot**
Distribution summary and unusual marks.

Average Line versus Target Line

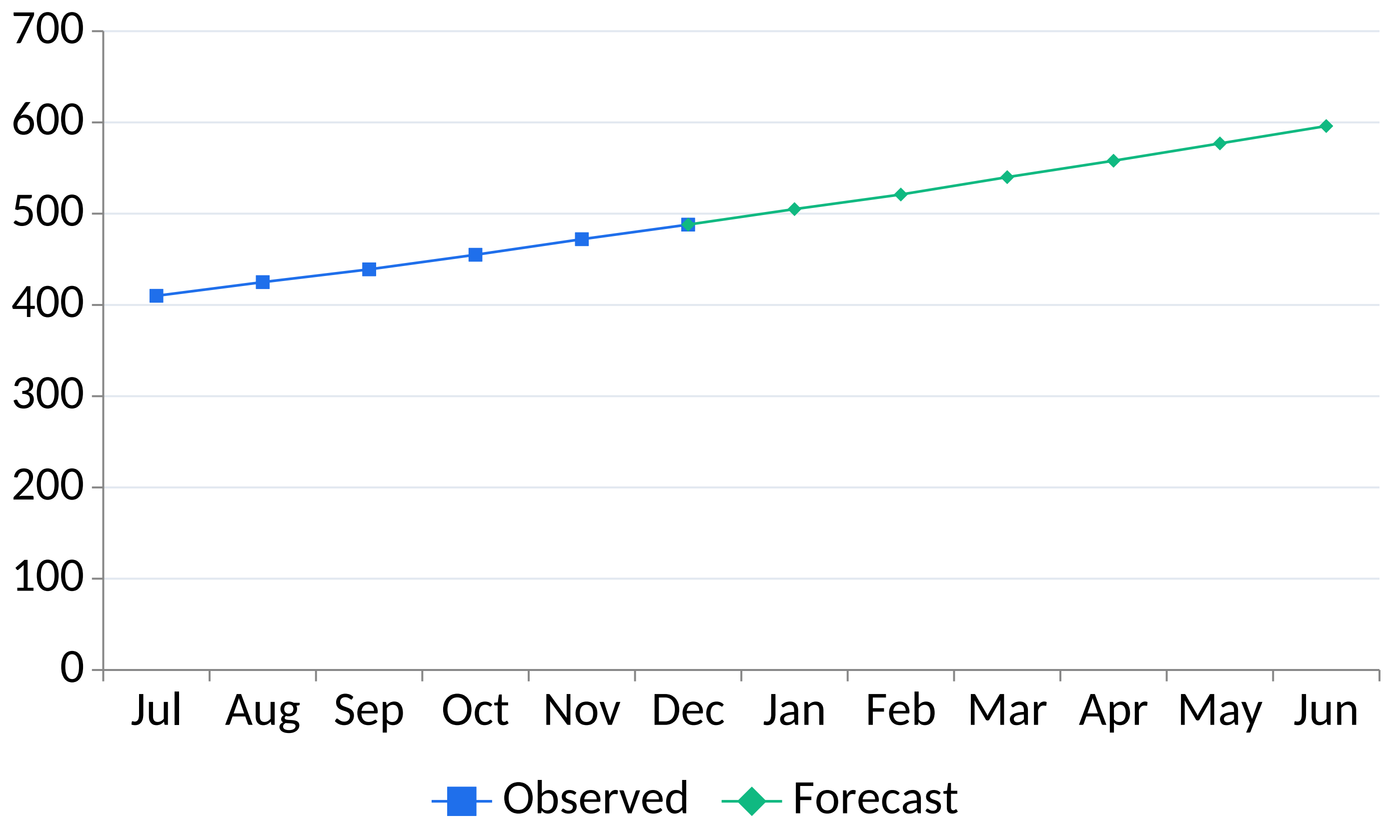
OBSERVED

1. Average summarises the observed population.
2. It moves when filters or the population change.
3. Useful for relative comparison.

EXPECTED

1. Target represents policy, plan or benchmark.
2. It should have an owner, period and source.
3. Do not label an average as expected performance.

Observed Trend and Forecast Context



READ THE EVIDENCE

The forecast extends the recent pattern, but the decision should use its prediction interval and monitor error after each refresh.

Editable chart: change the embedded data, labels, axes and colours directly in PowerPoint.

Trend Lines

EVIDENCE

1. Describe direction and fitted relationship.
2. Model form and level of detail affect the line.
3. R-squared summarises fit in the displayed sample.

LIMIT

1. Association is not causation.
2. Outliers and clusters can dominate the fit.
3. Inspect residual patterns and domain plausibility.

TOPIC 06 · K1 · A6 · LO6

Forecast Design

1 **Time grain**
Consistent intervals with enough history.

2 **Horizon**
Short enough for assumptions to remain plausible.

3 **Seasonality**
Recurring pattern supported by complete cycles.

4 **Prediction interval**
Range of uncertainty, not decorative shading.

5 **Missing values**
Treatment can change the model.

6 **Monitoring**
Compare forecast with actuals and refresh.

Clustering Workflow

Select features

- Decision-relevant measures
- Comparable scaling and meaning

Inspect profiles

- Cluster centres and membership
- Outliers and stability

Name cautiously

- Business labels after review
- Do not stereotype individuals

Interpretation Ladder

1. Observation
2. Pattern
3. Possible explanation
4. Limitation / alternative
5. Action and monitoring

Common Data Limitations

1. **Coverage**
   Period, channel or population may be incomplete.
2. **Quality**
   Missing, duplicated or inconsistent values.
3. **Granularity**
   Aggregation can conceal within-group variation.
4. **Bias**
   Selection or measurement process shapes the result.
5. **Drift**
   Historical relationships may change.
6. **Confounding**
   A third factor may explain an apparent relationship.

Correlation and Causation

CAN SAY

1. Correlation describes co-movement.
2. Trend lines estimate association under assumptions.
3. Useful for exploration and prediction.

CANNOT CLAIM

1. Causation needs design and alternative explanations.
2. Reverse causality and confounding remain possible.
3. Use cautious language in recommendations.

Responsible Data Story

Evidence

- What the visual directly shows
- Population, period and unit

Interpretation

- Most plausible business meaning
- Alternative explanations

Action

- Owner, trigger and next check
- Uncertainty and guardrail

Analytical Review Questions

1. **Right question**
   Does the view address a real decision?

2. **Right population**
   Do filters and exclusions match the claim?

3. **Right method**
   Is the visual/model suitable for the data?

4. **Right context**
   Are benchmarks, units and time visible?

5. **Right caveat**
   Are material limitations explicit?

6. **Right action**
   Can someone act and monitor the outcome?

TOPIC 06 · K1 · A6 · LO6

The strongest visual is honest about what it cannot prove.

Communicate the signal, the uncertainty and the next evidence needed for a better decision.

LAB 10 · LO6 · K1 · A6

Analyse Trends, Forecasts and Clusters

LAB 10

The strategy team needs a cautious outlook for outlet demand and a segmentation of store operating patterns.

EXPECTED OUTPUT

An analytics workbook with reference context, trend, forecast, clustering and a limitations-led decision narrative.

TOOLS Reference lines · trend lines · forecast · clustering · describe trend/model

Lab 10 Acceptance Evidence

ACCEPTANCE TEST

The workbook states assumptions and limitations, distinguishes association from causation, and reports at least one action with a monitoring trigger.

Build

Packaged Tableau workbook
Required views and controls

Prove

Screenshot with filter state
Excel reconciliation check

Explain

Observation and meaning
Limitation and next action

TOPIC RECAP

Recap — Analytics and Communication

1 **Knowledge**
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K1 · A6 · LO6

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WRAP-UP

Evidence, Assessment and Next Steps

Turn visual analysis into a defensible decision.

LEARNING OUTCOMES

What You Achieved

1 **LO1**
Apply basic Tableau features for displaying information.

2 **LO2**
Synthesise various plots to present data visually.

3 **LO3**
Transform data to create informative and dynamic data displays.

4 **LO4**
Create dashboards and scorecards to display internal and external benchmark data.

5 **LO5**
Apply calculations and parameters to create interactive graphics, visuals and technical features.

6 **LO6**
Perform analytics and communicate data limitations and interpretations of findings.

Courseware and Assessment on the LMS

Courseware

- Learner slide PDF
- Detailed Learner Guide
- Ten individual lab folders

Assessment

- WA question paper
- PP question paper
- Upload your own evidence

https://lms-tms.tertiaryinfotech.com/

Support

- Check the current TGS code
- Use the current release
- Contact trainer or academy

PRACTICE · SELF-CHECK

Tableau Practice Exam

Purpose

- Check concept readiness before assessment
- Use results to target revision

Focus

- Data roles, modelling and chart choice
- Dashboards, calculations and responsible interpretation

https://exams.tertiaryinfotech.com/

Readiness

- Complete individually
- Review explanations, then revisit the relevant topic and lab

Assessment Reminder

WA

- 5 open-ended knowledge questions
- 60 minutes

PP

- 6 connected Tableau tasks
- 60 minutes

Submission

- Complete assessment attendance
- Upload papers and required evidence

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then PP
4. Submit on the LMS
5. Sign Assessment Summary Record

Digital Attendance (Mandatory)

1 **Assessment**
Scan the SSG QR code before starting.

2 **Training**
Confirm AM and PM records for both days.

3 **Minimum**
At least 75% attendance is required.

4 **Support**
Tell the trainer immediately if attendance submission fails.

THANK YOU

Make the evidence easy to see — and hard to misunderstand.

Connect, model, visualise, interact, analyse and communicate with explicit limits.